

FasterCap 2D Input Files

Conductor definitions ('C' statement)

Syntax: C <file> <outperm> <xoffset> <yoffset> [+]

The 'C' element at the beginning of a line defines a conductor. The geometry of the conductor surface is further specified in the file <file>. The conductor is assumed embedded in an homogeneous dielectric medium with relative permittivity <outperm>. The relative permittivity <outperm> can be complex valued, in the format *ere-jeim*, where *ere* is the real part of the complex permittivity value, and *eim* is the imaginary part. The conductor can be translated with respect to the coordinates defined in <file> by an offset (*xoffset*, *yoffset*), thus allowing to reuse the same geometric definitions multiple times. The optional + argument is used to merge the current conductor with the next one, thus treating them a single conductor. With this option, panels with the same conductor name in different files will belong to the same conductor (see panels definitions for more information about conductor naming).

Example 1:

```
C circle_0.2_fine.txt 1.0 0.0 0.0
```

This input file fragments specifies a conductor whose geometry is defined in the file `circle_0.1_fine.txt`, embedded in a dielectric with permittivity equal to one (e.g. air), and translated to the position (0.0, 0.0).

Example 2:

```
C microstrip_top_0.03.txt 1.0 -0.025 0.01 +
C microstrip_bottom_0.03.txt 2.0 -0.025 0.01
```

This input file fragments specifies a conductor made of two parts in contact with two dielectric materials with different relative permittivities. It is considered a single conductor since the first statement ends by +, thus collating the two elements; i.e. the two parts are considered short-circuited together.

Example 3:

```
C circle_0.005.txt 4.0-j3.016e6 0.0 0.02
```

This input file fragments specifies a conductor whose geometry is defined in the file `circle_0.005.txt`, embedded in a dielectric with complex relative permittivity equal to 4.0-j3.016e6 (e.g. with real part equal to 4.0 and imaginary part equal to 3.016e6), and translated to the position (0.0, 0.02).